1(a)	$\left(2-\frac{x}{4}\right)^{10}=2^{10}+\binom{10}{1}2^9\left(-\frac{x}{4}\right)+\binom{10}{2}2^8\left(-\frac{x}{4}\right)^2+\binom{10}{3}2^7\left(-\frac{x}{4}\right)^3+\ldots$	M1
	$=1024-1280x+720x^2-240x^3$	B1 A1 A1
		(4)
(b)	$\left(3-\frac{1}{x}\right)^2=9-\frac{6}{x}+\frac{1}{x^2}$ or $9-\frac{3}{x}-\frac{3}{x}+\frac{1}{x^2}$	B1
	$\left(3-\frac{1}{x}\right)^2\left(2-\frac{x}{4}\right)^{10}=\left(9-\frac{6}{x}+\frac{1}{x^2}\right)\left(1024-1280x+720x^2\left(-240x^3\right)+\ldots\right)$ constant term $=9\times1024-\frac{6}{x}\left(-1280x\right)+\frac{1}{x^2}720x^2$	M1
	$=17616$	A1
		(3)
		Total 7

2(a)							
	x	-0.25	0	0.25	0.5	0.75	B1
	y	0.462	**0.577**	0.653	**0.686**	0.698	
							(1)
(b)	$h = 0.25$						B1
	$A \approx \frac{1}{2} \times "0.25" \left\{ 0.462 + 0.698 + 2\left("0.577" + 0.653 + "0.686" \right) \right\}$ Correct application of the trapezium rule with their h Must be a correct application of the rule so e.g. $A \approx \frac{1}{2} \times "0.25" \times 0.462 + 0.698 + 2\left("0.577" + 0.653 + "0.686" \right)$ Scores M0 unless any missing brackets are implied by subsequent work. $A \approx \frac{1}{2} \times "0.25" \left\{ 0.462 + 0.698 + 2\left("0.577" + 0.653 + "0.686" \right)$ Would also score M0 unless the closing bracket was implied by subsequent work						M1
	= awrt 0.624 or $\frac{78}{125}$ oe e.g. $\frac{312}{500}$						A1
							(3)
							Total 4

3(a)	$a(-4)^3-(-4)^2+b(-4)+4=-108$	M1
	$-64a-16-4b+4=-108$ $\Rightarrow 16a+b=24$ *	A1*
		(2)
(b)	$a\left(\frac{1}{2}\right)^3-\left(\frac{1}{2}\right)^2+b\left(\frac{1}{2}\right)+4=0$	M1
	$16a+b=24,\ a+4b=-30$ $\Rightarrow a=...,b=...$	M1
	$a=2,\ b=-8$	A1
		(3)
(c)	$\mathrm{f}(x)=2x^3-x^2-8x+4$ $\Rightarrow \mathrm{f}'(x)=6x^2-2x-8$	B1ft
		(1)

(d)	$6x^2-2x-8=0$ $\Rightarrow (3x-4)(x+1)=0$ $\Rightarrow x=...$		M1
	$x=\frac{4}{3}, -1 \Rightarrow y=...$		M1
	$\left(\frac{4}{3}, -\frac{100}{27}\right)$ **or** $(-1, 9)$ Or e.g. $x=\frac{4}{3},\ y=-\frac{100}{27}$ and $x=-1,\ y=9$ One correct point. The fractional coordinates must be exact but allow 1.3 with a dot over the		A1
	$\left(\frac{4}{3}, -\frac{100}{27}\right)$ **and** $(-1, 9)$ Or e.g. $x=\frac{4}{3},\ y=-\frac{100}{27}$ and $x=-1,\ y=9$ Both correct points. The fractional coordinates must be exact but allow 1.3 with a dot over the		A1
	Fully correct answers with no working scores 4/4 following a <u>correct</u> part (c) i.e. $\Rightarrow f'(x)=6x^2-2x-8$		
			(4)
			Total 10

4(a)(i)	(–7, 9) or e.g. $x = -7, y = 9$	B1 B1
	Award the marks in (a) once correct answers are seen. **Special case:** If all you see is (9, –7) award B1B0	
(a)(ii)	Examples: $r = \sqrt{\left(-3-\left("-7"\right)\right)^2+\left(12-"9"\right)^2}$ **or** $r = \sqrt{\left(-11-\left("-7"\right)\right)^2+\left(6-"9"\right)^2}$ **or** $r = \frac{1}{2}\sqrt{\left(-3+11\right)^2+\left(12-6\right)^2}$	M1
	$r = 5$	A1
		(4)
(b)	$\left(x+7\right)^2+\left(y-9\right)^2=5^2$ or e.g. $x^2+y^2+2\times7x-2\times9y+7^2+9^2-5^2=0$ M1: Correct attempt at circle equation using their values.	M1A1
		(2)
(c)	$m_{radius} = \frac{12-9}{-3+7}\left(=\frac{3}{4}\right)$ **or** $m_{tangent} = -1\div\left(\frac{12-9}{-3+7}\right)\left(=-\frac{4}{3}\right)$ **or** $m_{tangent} = -\left(\frac{-3+7}{12-9}\right)\left(=-\frac{4}{3}\right)$	M1
	Alternative for the first M: $\left(x+7\right)^2+\left(y-9\right)^2=5^2 \Rightarrow 2\left(x+7\right)+2\left(y-9\right)\frac{\mathrm{d}y}{\mathrm{d}x}=0$ $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{x+7}{9-y}=\frac{-3+7}{9-12}\left(=-\frac{4}{3}\right)$	
	$y-12=-\frac{4}{3}\left(x+3\right)$ Uses a correct straight line method for the **tangent** using the point Q. Must be fully correct work here so must be a clear attempt at the tangent not the radius. So if the radius gradient is found previously, must apply negative reciprocal rule to their radius gradient. If using $y = mx + c$ must reach as far as $c = \ldots$	M1
	$4x+3y-24=0$	A1
		(3)
		Total 9

5(a)	$t_{40} = 100 + (40-1)\times 5$	M1
	= (£)295	A1
		(2)
(b)	$S_{60} = \frac{1}{2}(60)(2\times 100 + (60-1)\times 5)$ or $l = 100 + (60-1)\times 5 = 395$ $S_{60} = \frac{1}{2}(60)(100+395)$	M1 A1
	= (£)14 850	A1
		(3)
(c)	$\frac{1}{2}n(2\times 600 + (n-1)\times -10) = 18200$	M1 A1
	$600n - 5n^2 + 5n = 18200$ $5n^2 - 605n + 18200 = 0$ $n^2 - 121n + 3640 = 0$ *	A1*
		(3)
(d)	$(n-56)(n-65) = 0$ $\Rightarrow (n=)56, 65$	M1 A1
		(2)
(e)	E.g. (n =) 65 because e.g. the money has already been saved after 56 months	B1
		(1)
		Total 11

6(a)	$x^3-6x+9=-2x^2+7x-1$ $\Rightarrow ...$	M1
	$\Rightarrow \pm\left(x^3+2x^2-13x+10\right)=0$	A1
	Examples: $x^3+2x^2-13x+10=\left(x-1\right)\left(x^2+...x+...\right)=(x-1)(x+...)(x+..)\Rightarrow x=...$	M1
	$x=2,\ y=5$ or $\left(2,\ 5\right)$ Correct values **from a correct cubic.**	A1
	Special Case If you see: $x^3-6x+9=-2x^2+7x-1\Rightarrow x^3+2x^2-13x+10=0$ $\Rightarrow x=2,\ y=5$ or $\left(2,\ 5\right)$ Score M1A1B1(**Second M on EPEN**)A0	
		(4)

(b)	$x^n \rightarrow x^{n+1}$	M1
	$\pm\int\left\{-2x^2+7x-1-\left(x^3-6x+9\right)\right\}\mathrm{d}x = \pm\int\left(-x^3-2x^2+13x-10\right)\mathrm{d}x$ $= \pm\left(-\frac{x^4}{4}-\frac{2x^3}{3}+\frac{13x^2}{2}-10x\right)$ or $\pm\left\{\int\left(-2x^2+7x-1\right)\mathrm{d}x-\int\left(x^3-6x+9\right)\mathrm{d}x\right\}$ $= \pm\left(-\frac{2x^3}{3}+\frac{7x^2}{2}-x-\left(\frac{x^4}{4}-\frac{6x^2}{2}+9x\right)\right)$ or $\int\left(-2x^2+7x-1\right)\mathrm{d}x = -\frac{2x^3}{3}+\frac{7x^2}{2}-x,\ \int\left(x^3-6x+9\right)\mathrm{d}x = \frac{x^4}{4}-\frac{6x^2}{2}+9x$ dM1: For correct integration of 1 term for C_1 **and** one term for C_2 or for correct integration for 2 terms of <u>their</u> $\pm(C_1 - C_2)$ A1: Fully correct integration of both C_1 and C_2 or for $\pm(C_1 - C_2)$. **Award this mark as soon as fully correct integration is seen and ignore subsequent work.**	dM1A1
	$= -\frac{2^4}{4}-\frac{2(2)^3}{3}+\frac{13(2)^2}{2}-10(2)-\left(-\frac{1^4}{4}-\frac{2(1)^3}{3}+\frac{13(1)^2}{2}-10(1)\right)$ Fully correct strategy for the area. Depends on both previous M marks. Uses the limits "2" and 1 in their "changed" expression(s) and subtracts either way round.	ddM1
	$= \frac{13}{12}$ If the attempt is correct apart from subtracting the wrong way round (for limits or functions) and $-\frac{13}{12}$ is obtained, allow recovery if they then make their answer positive.	A1
		(5)
		Total 9

7(i)	$\tan\theta+\frac{1}{\tan\theta}=\frac{\sin\theta}{\cos\theta}+\frac{1}{\frac{\sin\theta}{\cos\theta}}$ or $\tan\theta+\frac{1}{\tan\theta}=\frac{\sin\theta}{\cos\theta}+\frac{\cos\theta}{\sin\theta}$		M1
	$\frac{\sin\theta}{\cos\theta}+\frac{\cos\theta}{\sin\theta}=\frac{\sin^2\theta+\cos^2\theta}{\sin\theta\cos\theta}$ or $\frac{\sin^2\theta}{\sin\theta\cos\theta}+\frac{\cos^2\theta}{\sin\theta\cos\theta}$ Uses $\tan\theta\equiv\frac{\sin\theta}{\cos\theta}$ and $\frac{1}{\tan\theta}\equiv\frac{\cos\theta}{\sin\theta}$ and attempts common denominator of $\sin\theta\cos\theta$ with a 2 term numerator one of which is correct. Or attempts 2 separate fractions with a denominator of $\sin\theta\cos\theta$ one of which is correct. **Depends on the first mark.**		**d**M1
	$=\frac{1}{\sin\theta\cos\theta}$* or $\frac{1}{\cos\theta\sin\theta}$		A1*
			(3)
	Alternative 1 for (i)		
	$\tan\theta+\frac{1}{\tan\theta}=\frac{\tan^2\theta+1}{\tan\theta}\left(\text{or}\ \frac{\tan^2\theta}{\tan\theta}+\frac{1}{\tan\theta}\right)$		M1
	$=\frac{\sec^2\theta}{\tan\theta}=\frac{1}{\cos^2\theta}\times\frac{\cos\theta}{\sin\theta}$ Or $=\frac{\frac{\sin^2\theta}{\cos^2\theta}+1}{\frac{\sin\theta}{\cos\theta}}=\frac{1}{\cos^2\theta}\times\frac{\cos\theta}{\sin\theta}$		**d**M1
	$=\frac{1}{\sin\theta\cos\theta}$* or $\frac{1}{\cos\theta\sin\theta}$		A1*
	Alternative 2 for (i)		
	$\tan\theta+\frac{1}{\tan\theta}=\frac{1}{\sin\theta\cos\theta}\Rightarrow\frac{\sin^2\theta}{\cos\theta}+\cos\theta=\frac{1}{\cos\theta}$ Uses $\tan\theta\equiv\frac{\sin\theta}{\cos\theta}$ and multiplies through by $\sin\theta$ or $\cos\theta$		M1
	$\Rightarrow\sin^2\theta+\cos^2\theta=1$ Uses $\tan\theta\equiv\frac{\sin\theta}{\cos\theta}$ and multiplies through by $\sin\theta$ and $\cos\theta$		**d**M1
	$\sin^2\theta+\cos^2\theta=1$ is true hence proved		A1*

(ii)	$3\cos^2(2x+10°)=1 \Rightarrow \cos^2(2x+10°)=\frac{1}{3} \Rightarrow \cos(2x+10°)=(\pm)\sqrt{\frac{1}{3}}$ Divides by 3 and takes square root **<u>of both sides</u>**. The "±" is not required.	M1
	$2x+10°=\cos^{-1}\left("(\pm)\sqrt{\frac{1}{3}}"\right)$ $\Rightarrow x=\frac{\cos^{-1}\left("(\pm)\sqrt{\frac{1}{3}}"\right)-10°}{2}$	M1
	$x=22.4°$ **or** $x=57.6°$	A1
	$x=22.4°$ **and** $x=57.6°$	A1
	If mixing degrees and radians allow the method marks.	
		(4)
	Alternative 1 for part (b)	
	$3\cos^2(2x+10°)=1 \Rightarrow 3\left(1-\sin^2(2x+10°)\right)=1 \Rightarrow$ $\Rightarrow \sin^2(2x+10°)=\frac{2}{3} \Rightarrow \sin(2x+10°)=(\pm)\sqrt{\frac{2}{3}}$ Uses a correct identity, rearranges and takes square root **<u>of both sides</u>**. The "±" is not required.	M1
	$2x+10°=\sin^{-1}\left("(\pm)\sqrt{\frac{2}{3}}"\right)$ $\Rightarrow x=\frac{\sin^{-1}\left("(\pm)\sqrt{\frac{2}{3}}"\right)-10°}{2}$	M1
	$x=22.4°$ **or** $x=57.6°$	A1
	$x=22.4°$ **and** $x=57.6°$	A1
	Alternative 2 for part (b)	
	$3\cos^2(2x+10°)=3\left(\frac{1+\cos(4x+20)}{2}\right) \Rightarrow \cos(4x+20)=-\frac{1}{3}$ Uses a correct identity, rearranges to make $\cos(4x+20)$ the subject	M1
	$2x+10°=\cos^{-1}\left("-\frac{1}{3}"\right)$ $\Rightarrow x=\frac{\cos^{-1}\left("-\frac{1}{3}"\right)-20°}{4}$	M1
	$x=22.4°$ **or** $x=57.6°$	A1
	$x=22.4°$ **and** $x=57.6°$	A1
		Total 7

8(a)	$S_n = a + ar + ... + ar^{n-1}$ $rS_n = ar + ar^2 + ... + ar^n$ Writes down at least 3 correct terms of a geometric series and multiplies their sequence by r. There may be extra incorrect terms but allow this mark if there are 3 correct terms in both sequences and at least one "+" in both sequences **but see special case below**	M1
	$S_n - rS_n = a - ar^n$ or $rS_n - S_n = ar^n - a$ Obtains either equation where both S_n and rSn had the correct first and last terms and at least one other correct term but no incorrect terms. Both sides must be seen unfactorised.	A1**(M1 on EPEN)**
	$(1-r)S_n = a(1-r^n) \Rightarrow S_n = \frac{a(1-r^n)}{1-r}$ * Factorises both sides and divides by $1 - r$ to obtain the printed answer Should be as printed but allow e.g. $S_n = \frac{a(1-r^n)}{(1-r)}$ but not $S_n = \frac{a(r^n-1)}{(r-1)}$ unless followed by correct version	A1*
	Special case: If terms are listed rather than added and the working is otherwise correct score 110 **See next page for proof by induction.**	
		(3)
	Alternative for (a):	
	$S_n = a + ar + ... + ar^{n-1}$ $(1-r)S_n = (1-r)(a + ar + ... + ar^{n-1})$ or $S_n = \frac{(1-r)(a + ar + ... + ar^{n-1})}{(1-r)}$ Writes down at least 3 correct terms of a geometric series and multiplies both sides by $1 - r$ or multiplies the right hand side by $\frac{1-r}{1-r}$ There may be extra incorrect terms but allow this mark if there are 3 correct terms	M1
	$(1-r)S_n = a - ar^n$ or $S_n = \frac{a-ar^n}{1-r}$ Obtains the above equation where S_n had the correct first and last terms and at least one other correct term and no incorrect terms. Right hand side must be seen unfactorised unless the "a" was factored out earlier	A1 **(M1 on EPEN)**
	$(1-r)S_n = a - ar^n = a(1-r^n) \Rightarrow S_n = \frac{a(1-r^n)}{1-r}$ * or $S_n = \frac{a-ar^n}{1-r} \Rightarrow S_n = \frac{a(1-r^n)}{1-r}$ * Should be as printed but allow e.g. $S_n = \frac{a(1-r^n)}{(1-r)}$ but not $S_n = \frac{a(r^n-1)}{(r-1)}$ unless followed by correct version	A1*

(b)	**Mark (b) and (c) together**	
	$r^3 = -\frac{20.48}{320} \Rightarrow r = \sqrt[3]{-\frac{20.48}{320}}$	M1
	$= -0.4$	A1
		(2)

(c)	$r = -0.4 \Rightarrow a = \dfrac{-320}{-0.4}\left(=800\right)$ or $r = -0.4 \Rightarrow a = \dfrac{512}{25} \div \left(-\dfrac{2}{5}\right)^4 \left(=800\right)$	M1
	$S_{13} = \dfrac{"800"\left(1-"-0.4"^{13}\right)}{1-"-0.4"}$ Correct attempt at the sum using their a and **their** r and $n = 13$ to find a value for S_{13}. **Must be a fully correct attempt at the sum here using $n = 13$, their a and their r.**	M1
		(3)
		Total 8

9(i)	$4=\log_3 81$ or $4=\log_3 3^4$	B1
	Examples: $\log_3(x+5)-\log_3 81=\log_3\dfrac{x+5}{81}$ or $\log_3(x+5)-\log_3(2x-1)=\log_3\dfrac{x+5}{2x-1}$ or $\log_3(2x-1)+\log_3 81=\log_3 81(2x-1)$	M1
	$\dfrac{x+5}{81}=2x-1$	A1
	$x=\dfrac{86}{161}$	A1
		(4)
	Alternative for first 3 marks: $\log_3(x+5)-4=\log_3(2x-1)\Rightarrow 3^{\log_3(x+5)-4}=3^{\log_3(2x-1)}$ $\Rightarrow 3^{\log_3(x+5)}\times 3^{-4}=2x-1\Rightarrow\dfrac{x+5}{81}=2x-1$ Score B1 for sight of 3^{-4} and M1 for applying $3^{a\pm b}=3^a\times 3^{\pm b}$ and A1 as above	
	(a) Special Case $\log_3(x+5)-\log_3(2x-1)=4\Rightarrow\dfrac{\log_3(x+5)}{\log_3(2x-1)}=4\Rightarrow\dfrac{x+5}{2x-1}=81\Rightarrow x=\dfrac{86}{161}$ Scores B1(implied) M0 A0 A1	
(ii)(a)	$3^{y+3}=3^y\times 3^3$ **or** $2^{1-2y}=2\times 2^{-2y}$ or $\dfrac{2}{2^{2y}}$ or 2×4^{-y} or $\dfrac{2}{4^y}$	B1
	$3^{y+3}\times 2^{1-2y}=27\times 3^y\times 2\times 2^{-2y}=\ldots$	M1**(B1 on EPEN)**
	$3^y\times 2^{-2y}=\dfrac{108}{27\times 2}$ or $\dfrac{3^y}{4^y}=\dfrac{108}{27\times 2}$ or $\dfrac{3^y}{2^{2y}}=\dfrac{108}{54}$ Isolates the terms in y (as powers of 3 and 2(or 4)) on the lhs and the constants on the rhs. There must be no incorrect work to combine terms e.g. $3^y\times 3^3=27^y$ etc.	M1
	$(0.75)^y=2$ *	A1*
		(4)

	Alternative 1 for (ii)(a) using logs:		
	$\log\left(3^{y+3}\times 2^{1-2y}\right)=\log 3^{y+3}+\log 2^{1-2y}$ Or $\log 3^{y+3}=(y+3)\log 3$ Or $\log 2^{1-2y}=(1-2y)\log 2$	One **correct** log law seen or implied anywhere in their working. **No bracketing errors allowed for <u>this mark</u>.**	B1
	$\log 3^{y+3}+\log 2^{1-2y}=(y+3)\log 3+(1-2y)\log 2$ Applies the correct log laws to the lhs.		M1**(B1 on EPEN)**
	$(y+3)\log 3+(1-2y)\log 2=\log 108\Rightarrow \log 3^{y}-\log 2^{2y}=\log 108-3\log 3-\log 2$ $\Rightarrow \log\frac{3^{y}}{2^{2y}}=\log\frac{108}{3^{3}\times 2}$ Proceeds to isolate the terms in y on the lhs and combines the constants on the rhs or e.g. $(y+3)\log 3+(1-2y)\log 2=\log 108\Rightarrow y(\log 3-2\log 2)=\log\frac{108}{3^{3}\times 2}$ Proceeds to isolate the terms in y on the lhs and combines the constants on the rhs		M1
	$(0.75)^{y}=2*$	Cso. With e.g. 2log 2 seen as log 4 or log 2^2 or implied at some point.	A1*
	Alternative 2 for (ii)(a) using factors of 108:		
	$3^{y+3}\times 2^{1-2y}=108=2^{2}\times 3^{3}$ $\Rightarrow \frac{3^{y+3}\times 2^{1-2y}}{2^{2}\times 3^{3}}=\ldots$ $\Rightarrow 3^{y}\times 2^{-1-2y}=\ldots$	One correct index law seen or implied anywhere in their working e.g. $\frac{3^{y+3}}{3^{3}}=3^{y}$ or $\frac{2^{1-2y}}{2^{2}}=2^{-1-2y}$	B1
	$\Rightarrow 3^{y}\times 2^{-1-2y}=\ldots$	Applies both correct index laws to the lhs of the equation	M1**(B1 on EPEN)**
	$\Rightarrow 3^{y}\times 2^{-2y}=2$	Proceeds to isolate the terms in y (as powers of 3 and 2(or 4)) on the lhs and the constants on the rhs	M1
	$(0.75)^{y}=2*$	Cso. Reaches the given answer with no errors and all steps shown with 2^{2y} appearing as 4^{y} at some point.	A1*
(b)	$(0.75)^{y}=2\Rightarrow y=\frac{\log 2}{\log 0.75}$	Correct processing to obtain a value for y	M1
	$y=-2.409$	Awrt -2.409 A correct answer implies both marks	A1
			(2)
			Total 10